

Unit 1

Understanding Large Language Model

Language AI

- **Language AI** refers to artificial intelligence systems that can **understand, generate, reason with, and interact using human language** by learning patterns from large-scale text data.
- **Language modeling (LM)** forms the backbone of natural language processing (NLP), enabling machines to understand, generate, and predict human language.
- At its core, a language model **calculates the probability of sequences of words (tokens), predicting the next token based on preceding context.**
- A language model (LM) is like a smart autocomplete system: it predicts what comes next in a sentence.
- example: "Mother said grandma was ill, so last Sunday, Jane and I visited ____."
- A robust LM predicts "her" by analyzing contextual relationships between "grandma," "ill," and "visit"
- It uses **conditional probabilities.**

Language Modeling

- “The weather is nice today” $P(\text{"The", "weather", "is", "nice", "today"}) = P(\text{"The"}) \times P(\text{"weather" | "The"}) \times P(\text{"is" | "The weather"}) \times \dots$ So that it can predict the next word in a sentence score how "natural" a sentence is generate coherent text by sampling word-by-word
- Input: “The weather today is _____”. A language model might output: “Sunny” — 8.7%, “Rainy” — 5.1%, “Unpredictable and humid” — 1.6%
- It generates the next word (or phrase) by selecting the option with the highest probability at each step, based on the input context.

Language AI

- Core Components of Language AI
 - ✓ **Language Modeling:** predicting next tokens in a sequence
 - ✓ **Natural Language Understanding (NLU):** entity recognition, sentiment analysis
 - ✓ **Natural Language Generation (NLG):** text generation, summarization, dialogue
 - ✓ **Reasoning & Context Handling:** multi-turn conversations, long-context understanding
- **Evolution of NLP**

Rule-based → Statistical → Neural → Transformer-based models->LLM

Evolution of NLP

Rule-based System

- Language understanding through hand-crafted linguistic rules.

Techniques:

- Grammar rules
- Dictionaries and lexicons
- Pattern matching (IF-THEN rules)
- Finite State Machines

Examples:

- Early machine translation systems
- ELIZA chatbot

Advantages:

- Interpretable and deterministic
- Works well in constrained domains

Limitations:

- Not scalable
- Cannot handle ambiguity
- Requires extensive human expertise

Evolution of NLP

Statistical NLP

- Language is modelled using probability and statistics from data.

Techniques:

- N-gram language models
- Hidden Markov Models (HMM)
- Bayesian models

Examples:

- POS tagging using HMM
- Speech recognition systems

Advantages:

- Data-driven
- Handles uncertainty better than rules

Limitations:

- Data sparsity
- Fixed context window
- Feature engineering required

Evolution of NLP

Neural NLP

- Learn distributed representations using neural networks.

Techniques:

- Word embeddings (Word2Vec, GloVe, FastText)
- Feedforward Neural Language Models
- RNN, LSTM, GRU
- CNNs for text classification

Examples:

- Neural machine translation
- Sentiment analysis

Advantages:

- Captures semantic relationships
- Handles variable-length input
- Less manual feature engineering

Limitations:

- Sequential processing (slow training)
- Difficulty with very long contexts
- Vanishing/exploding gradients

Evolution of NLP

Transformer-Based NLP

- Use self-attention to model global dependencies in parallel.

Key Breakthrough:

- “Attention Is All You Need” (Vaswani et al., 2017)

Techniques:

- Self-attention & multi-head attention
- Encoder–Decoder Transformers
- Large-scale pre-training

Models:

- BERT (encoder-only), GPT (decoder-only) , T5, BART (encoder–decoder), LLaMA, PaLM

Advantages:

- Parallel computation
- Long-range dependency modelling
- Scales to trillion-token training
- Foundation for LLMs

Limitations:

- High computational cost
- Energy consumption
- Data and bias issues

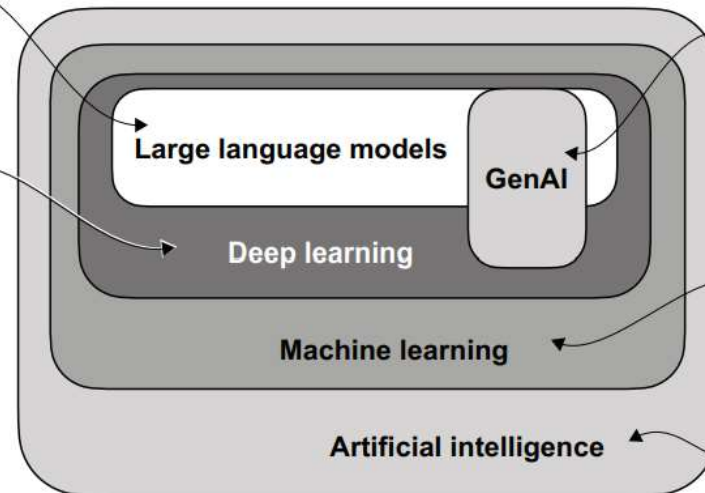
Large Language Model (LLM)

- An LLM is a **neural network designed to understand, generate, and respond to humanlike text.**
- The “Large” in “Large Language Model” refers to **both the model’s size in terms of parameters and the immense dataset on which it’s trained.**
- LLM have **tens or even hundreds of billions of parameters**, which are the adjustable weights in the network that are optimized during **massive training to predict the next word in a sequence.**
- Next-word prediction is sensible because it harnesses the inherent sequential nature of language to train models on **understanding context, structure, and relationships within text.**
- Since LLMs are capable of generating text, LLMs are also often referred to as a form of generative artificial intelligence, often abbreviated as **generative AI or GenAI.**

Large Language Model (LLM)

Deep neural network for parsing and generating human-like text

Machine learning with neural networks consisting of many layers



GenAI involves the use of deep neural networks to create new content, such as text, images, or various forms of media

Algorithms that learn rules automatically from data

Systems with human-like intelligence

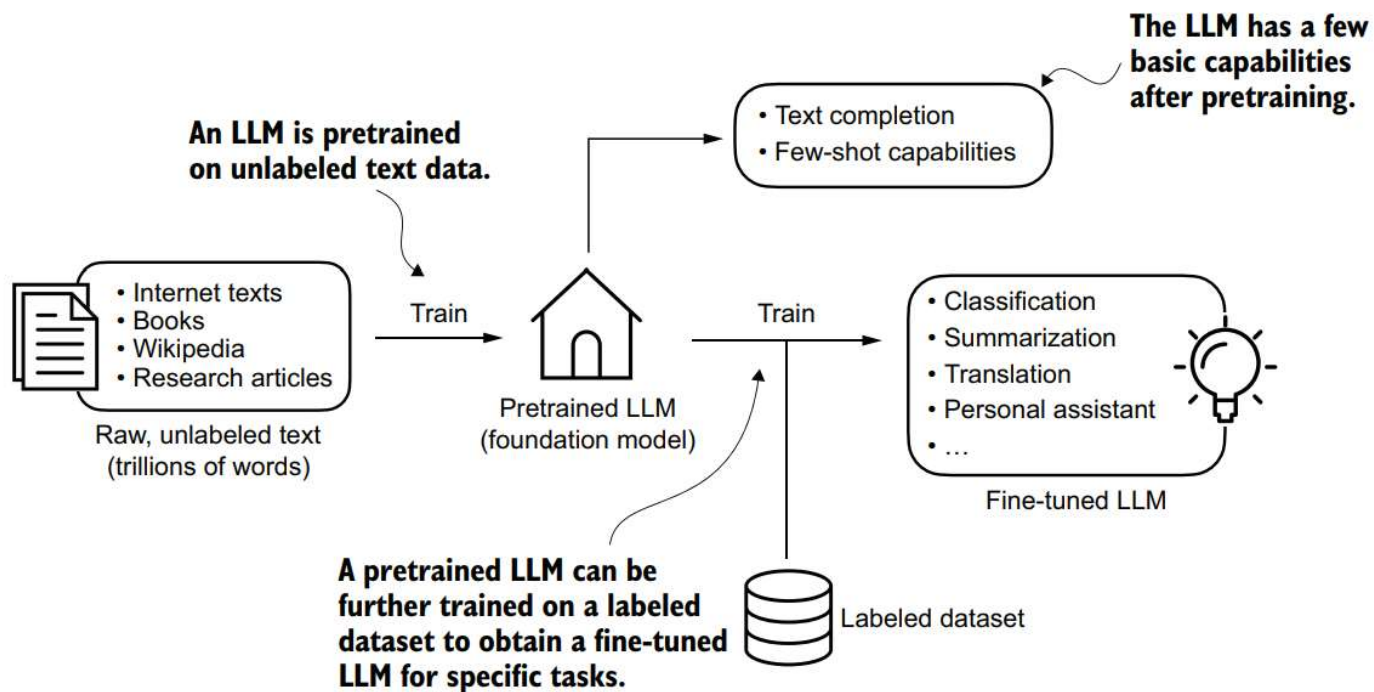
Large Language Model (LLM)

- **Applications of LLM**
- LLMs are employed for **machine translation, generation of novel texts, sentiment analysis, text summarization**, and many other tasks.
- It has been used for content creation, such as **writing fiction, articles, and even computer code**.
- LLMs can also power **sophisticated chatbots and virtual assistants**, such as OpenAI's ChatGPT or Google's Gemini
- It can be used for **effective knowledge retrieval** from vast volumes of text in specialized areas such as **medicine or law**. This includes sifting through documents, **summarizing lengthy passages, and answering technical questions**.

Large Language Model (LLM)

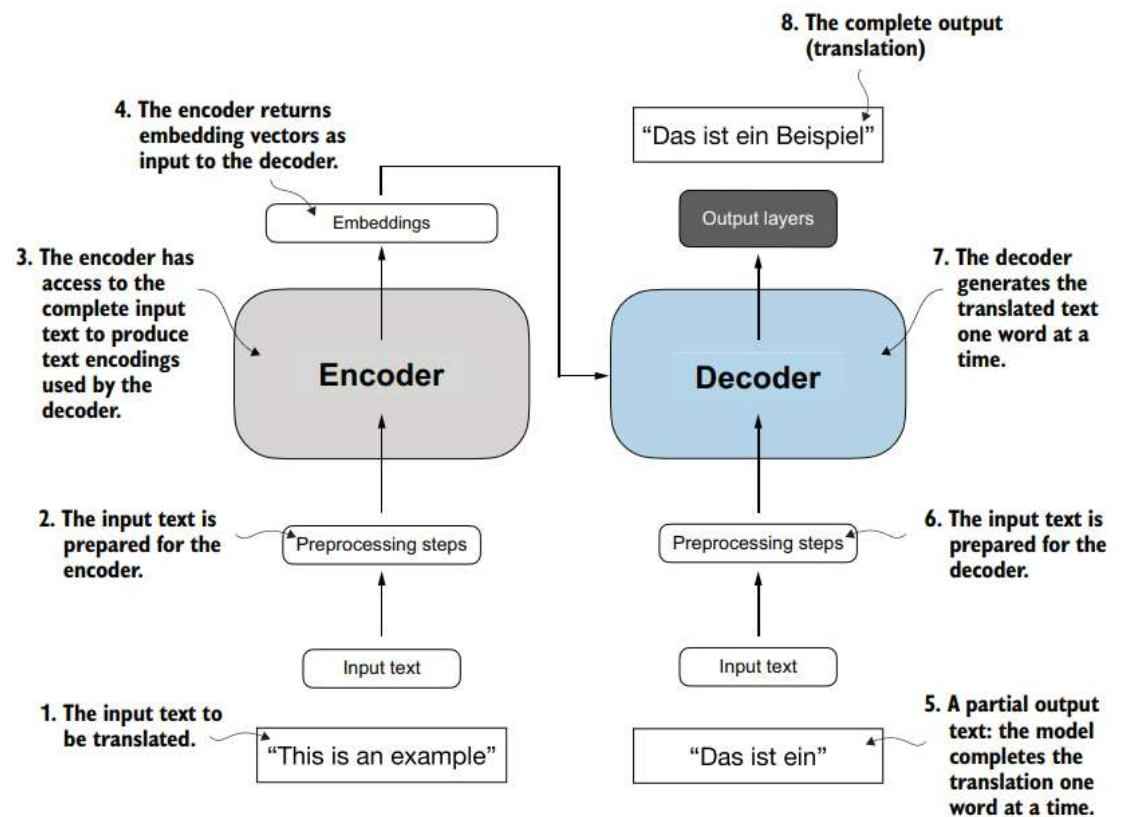
- **Stages of building and using LLMs**
- **Custom-built LLMs can outperform** general-purpose LLMs, such as those provided by ChatGPT. BloombergGPT (Specialized for finance). It also **provides privacy**. Additionally, developing smaller custom LLMs **enables deployment directly on customer devices**.
- The general process of creating an LLM includes **pretraining and fine-tuning**.
- The first step in creating an LLM is to **train it on a large corpus of text data**, sometimes referred to as raw text. (Unlabeled text).
- **Pretrained LLM** having capabilities of text completion is **trained on labeled dataset** to obtain finetuned LLM for specific task.
- Finetuned LLM is now able to **perform the dedicated task**.

Large Language Model (LLM)



Introducing the transformer architecture

- Most modern **LLMs** rely on the **transformer architecture**, which is a deep neural network architecture introduced in the 2017 paper “Attention Is All You Need”.
- Original transformer was developed for machine translation, translating English texts to German and French.

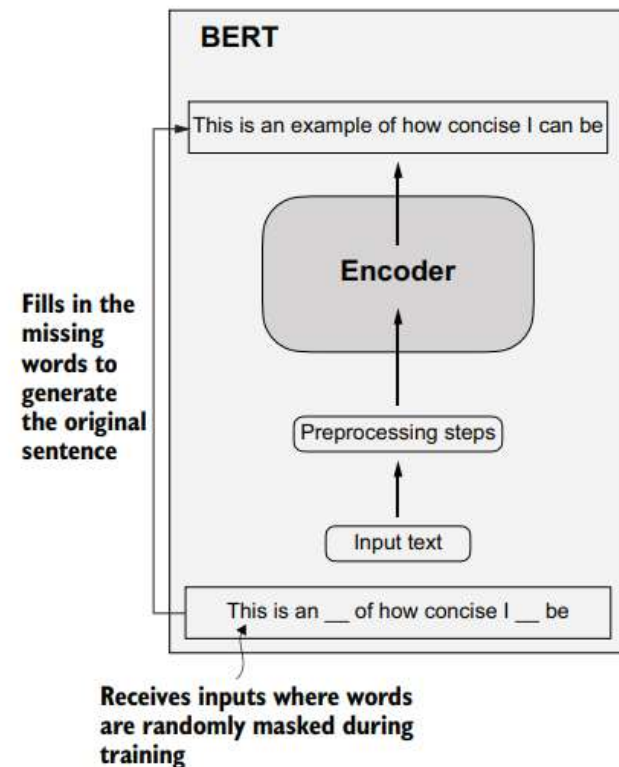


Introducing the transformer architecture

- A simplified depiction of the original transformer architecture, which is a **deep learning model for language translation**.
- The transformer consists of **two parts**:
 1. An **encoder** that processes the input text and **produces an embedding representation** (a numerical representation that captures many different factors in different dimensions) of the text that the
 2. **Decoder** can use to **generate the translated text one word at a time**. This figure shows the final stage of the translation process where the decoder has to generate only the final word (“Beispiel”), given the original input text (“This is an example”) and a partially translated sentence (“Das ist ein”), to complete the translation.
- Variants of the transformer architecture:
- BERT (Bidirectional Encoder Representations from Transformers)
- GPT (Generative Pretrained Transformers)

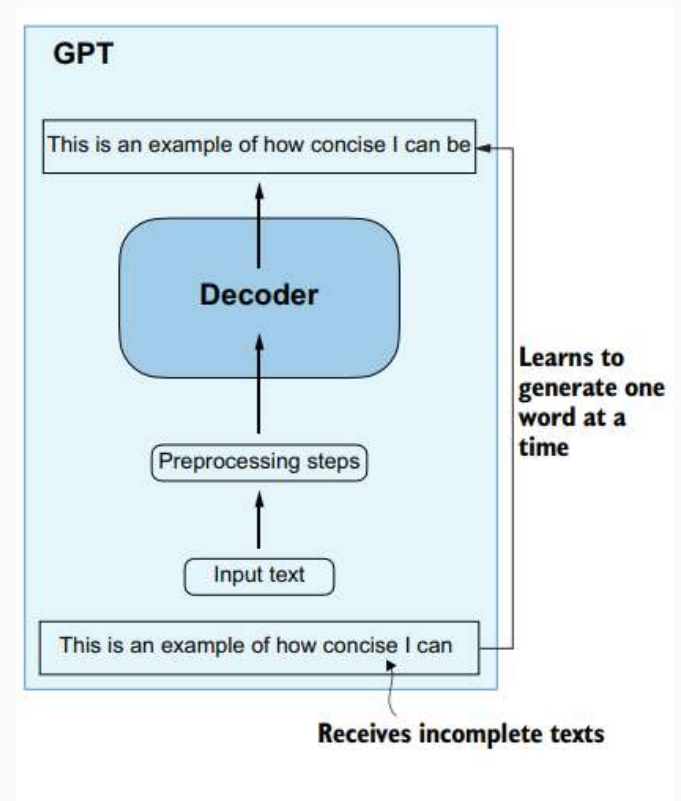
Introducing the transformer architecture

- **BERT Model**
- BERT is built upon the original **transformer's encoder submodule**
- BERT and its variants specialize in **masked word prediction**, where the model predicts masked or hidden words in a given sentence.
- This unique training strategy equips BERT with strengths in **text classification tasks**, including sentiment prediction and document categorization.
- As an application of its capabilities, **X** (formerly Twitter) uses BERT to **detect toxic content**.



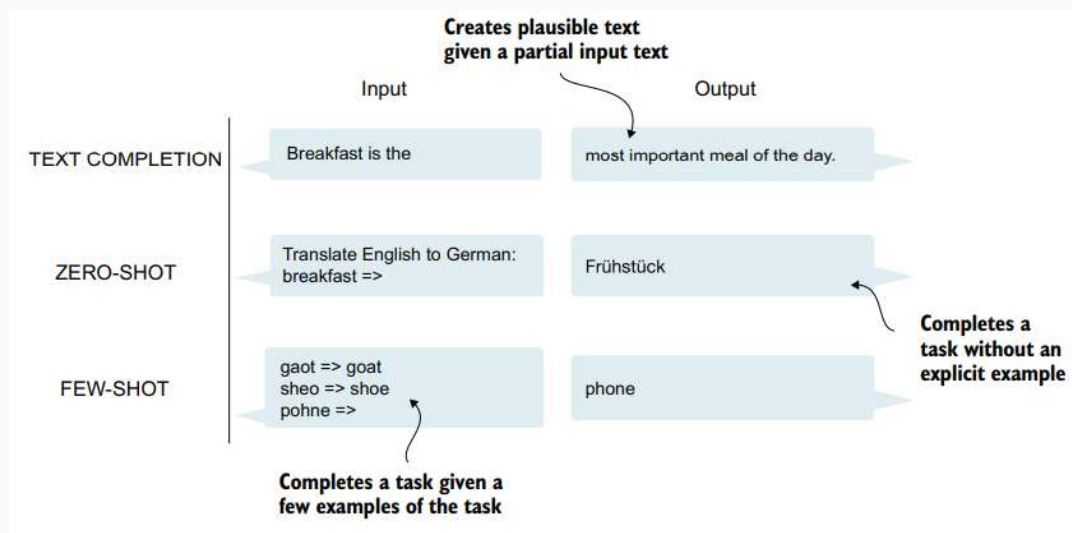
Introducing the transformer architecture

- **GPT Model**
- GPT focuses on the **decoder portion** of the original transformer architecture and is designed for tasks that **require generating texts**.
- This includes machine translation, text summarization, fiction writing, writing computer code, and more
- These models are adept at executing both zero-shot and few-shot learning tasks.



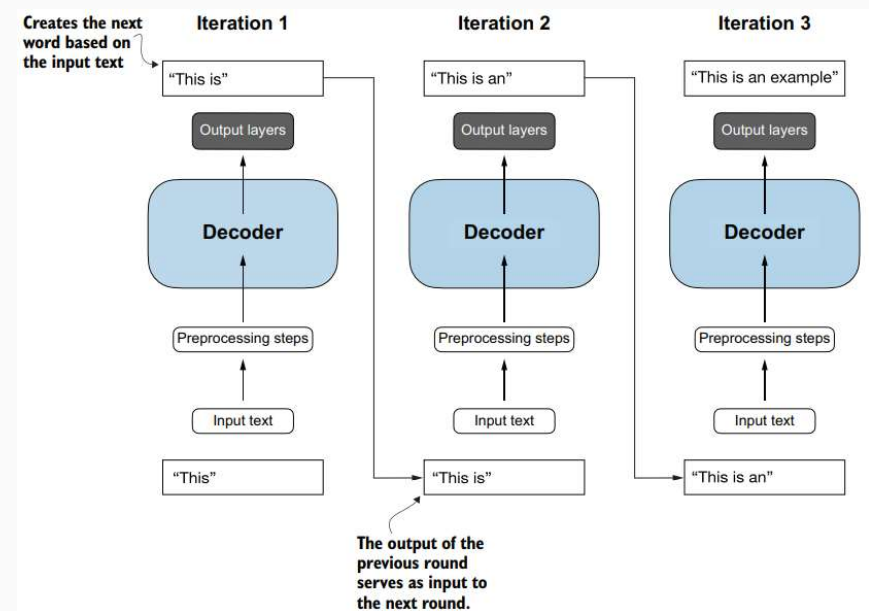
Introducing the transformer architecture

- **Zero-shot tasks.**
- Zero-shot learning refers to the ability to generalize to completely unseen tasks without any prior specific examples.
- **Few-shot learning tasks**
- Few-shot learning involves learning from a minimal number of examples the user provides as input



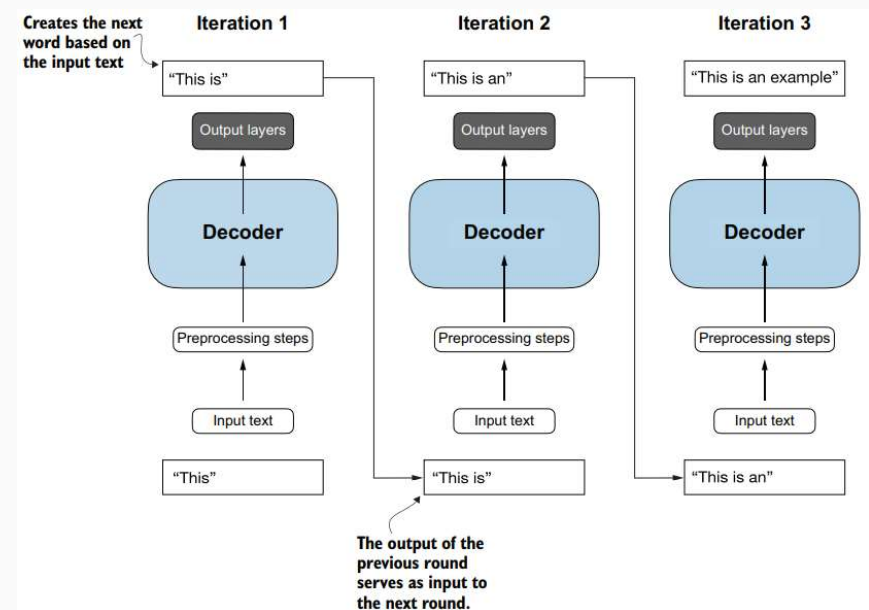
Introducing the transformer architecture

- Since decoder-style models like GPT generate text by predicting text one word at a time, they are considered a type of **autoregressive model**.
- Autoregressive models **incorporate their previous outputs as inputs for future predictions**.
- Consequently, in GPT, **each new word is chosen based on the sequence that precedes it**, which improves the coherence of the resulting text.

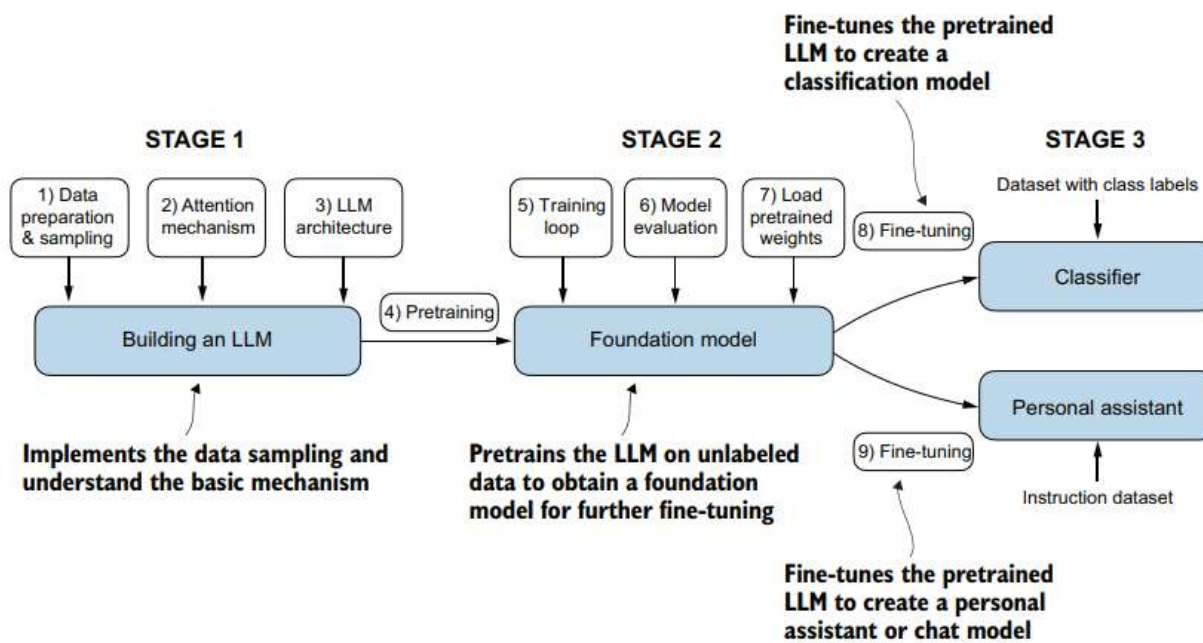


Introducing the transformer architecture

- The ability to perform tasks that the model wasn't explicitly trained to perform is called an **emergent behavior**.
- This capability **isn't explicitly taught** during training but emerges as a natural consequence of the model's exposure to vast quantities of multilingual data in diverse contexts.
- The fact that **GPT models can "learn" the translation patterns** between languages and perform translation tasks even though they **weren't specifically trained for it** demonstrates the benefits and capabilities of these largescale, generative language models.



Building LLM



Building LLM

- Stage 1 includes fundamental data preprocessing steps and the attention mechanism at the heart of every LLM.
- In stage 2, pretrains a GPT-like LLM capable of generating new texts. Pretraining an LLM from scratch is a significant endeavor, demanding thousands to millions of dollars in computing costs for GPT-like models.
- Stage 3 takes a pretrained LLM and fine-tune it to follow instructions such as answering queries or classifying texts—the most common tasks in many real-world applications and research.